



Chronic pain is specifically associated with updating working memory: a longitudinal twin study

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Abstract

Worse executive function (EF) is associated with chronic pain and could mechanistically contribute to pain chronification. It is unclear whether there is overall impairment in EFs or whether there are impairments in specific cognitive domains. Furthermore, the possible genetic risk underlying these associations has not been tested. Participants were from the Colorado Longitudinal Twin study; 786 same-sex twins completed a battery of EF tasks at ages 23 and/or 28 and 634 of these twins self-reported chronic pain at mean age = 28.1; prevalence = 27.76% using the Brief Pain History Questionnaire. The EF tasks were used to define a Common EF factor and 2 factors specific to updating working memory and shifting mental set. We estimated the phenotypic and genetic associations of stable EF variance across ages 23 and 28, as well as EF variance unique to age 28, with pain. With respect to stable EF variance, pain phenotypically correlated with the Updating-specific factor ($r = -0.21$, $P = 0.008$) but did not significantly correlate with the Common EF factor ($r = -0.06$, $P = 0.350$) nor with the Shifting-specific factor ($r = -0.03$, $P = 0.709$). There were no significant phenotypic correlations between pain and EF variance unique to age 28. A twin model indicated that pain and Updating-specific variance share genetic risk ($r_A = -0.46$, $P = 0.005$) but not environmental risk ($r_E = 0.05$, $P = 0.844$). Updating working memory shares a phenotypic and genetic relationship with pain in young adults. Impairments in gating or monitoring pain signals may play a mechanistic role in pain development.

Keywords: Genetics, Twin, Chronic pain, Executive function, Longitudinal, Latent variable

1. Introduction

Patients with chronic pain often show lower cognitive performance compared with individuals not in pain. In systematic reviews, individuals experiencing chronic pain perform worse across many cognitive tasks that tap various executive functions (EFs), including working memory, response inhibition, and cognitive flexibility.^{6,8,9,13} Pain relates to EFs throughout the lifespan, with young adolescents with pain exhibiting lower cognitive performance compared with their peers,²⁹ and older adults in pain showing steeper cognitive decline⁴² and increased risk for dementia.²² Prospectively, worse EFs predict pain incidence 9 years later, over and above various covariates that are related to pain, including psychopathology and socioeconomic status.⁴⁵ Preoperative patients with no pain before surgery who performed worse on cognitive tasks were more susceptible to postoperative pain chronification 6 and 12 months

postsurgery.³ Understanding the role of EFs in pain development could help clarify the pathophysiology of chronified pain.

Most of this previous research has tested associations with individual cognitive tasks. Cognitive tasks, particularly EF tasks, are susceptible to measurement error and often tap multiple cognitive processes.⁴⁰ Latent variable modeling allows for the extraction of shared variance across tasks meant to capture a specific construct (eg, working memory updating; mental set shifting), which minimizes measurement error. One study using this approach found that patients with pain had worse EF performance on a global EF factor constructed from 6 tasks⁵²; however, it remains unclear what EF domain(s) (eg, mental set shifting or updating working memory) are associated with pain. Prospective associations between EFs and pain may also differ when including multiple factors.

While poorer cognitive abilities during pain are supported in the literature, findings vary across tasks, and phenotypic studies cannot evaluate whether shared predispositions explain relationships. To our knowledge, no study has evaluated the genetic and environmental etiology of relationships between pain and EFs. Approximately 55% of the variance of pain appears to be due to genetic risk and remains stable across adulthood.¹⁴ EFs in adulthood are moderately to highly heritable (42%–100%) at the level of latent variables,^{22–25} and their genetic influences also remain stable across young adulthood.²⁵ These genetic influences on pain and EFs might overlap. There could also be overlapping environmental risk for pain and EFs, as both are sensitive to similar risk factors such as chronic metal exposure^{47,58} or stress.^{1,36}

In this twin study, we evaluated whether individual differences in 3 EF latent variables measured at ages 23 and 28 years (Common EF; working memory Updating-specific function; and

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mental set Shifting-specific function) are associated with pain reported at age 28. Then, we decomposed the significant associations into etiological correlations. To our knowledge, this is the first study to examine longitudinal and genetic relationships between 3 separable EF factors and pain. Examining associations with multiple EF factors elucidates which facets of EF (eg, Common EF) are specifically related to pain in young adults, and understanding the nature of these associations clarifies whether the pain–EF relationships are due to overlapping genetic and/or environmental risks.

2. Method

2.1. Participants

Participants were 786 individual same-sex twins (55% women, 45% men) from the Colorado Longitudinal Twin Study (LTS) with data for pain and/or EFs. Of the twins, 416 were monozygotic (MZ) and 370 were dizygotic (DZ). The LTS is a long-standing, ongoing developmental study of twins born between 1984 and 1990. Twins whose families lived within a 3-hour driving range (~300 km) from Boulder, Colorado, and whose birth weights were at least 1000 g (though 96% weighed 1700 g or more) were invited to participate (more details on data ascertainment are described in Refs. 16 and 50). Twins were not selected or excluded for particular conditions (eg, pain). The LTS began testing twins born in Colorado between 1986 and 1990 at age 14 months and followed them into adulthood.

For this study, we used data collected at 2 separate visits, when the twins were mean ages 22.84 (SD = 1.29, range = 21.11–28.03; total *n* = 785) and 28.80 (SD = 0.87, range = 28.02–34.58; total *n* = 649). EF data were collected at both visits, but the pain data were obtained only at the second time point of these visits when the twins completed functional magnetic resonance imaging and questionnaires. Exclusion criteria for this time point included standard contraindications to magnetic resonance imaging (metal in the body, recent tattoos, claustrophobia, and pregnancy).

The LTS sample self-reported their race as being 91.66% White, 4.44% Latino or Hispanic, 2.60% bi- or multiracial, <1% Pacific Islander, <1% American Indian, and <1% self-reporting as other. Fewer than 0.5% (*n* = 5) of the sample endorsed any usage (days 1–30) of painkillers in the last 30 days; it did not significantly predict pain. All study procedures were approved by the Institutional Review Board of the University of Colorado Boulder.

2.2. Measures

2.2.1. Brief Pain History Questionnaire

The Brief Pain History Questionnaire (BPHQ) is a self-report questionnaire based on the Brief Pain Inventory, which is a validated tool for assessing nonmalignant pain.⁵⁵ The BPHQ was completed by the participants at the age 28 assessment. The BPHQ asks participants to report whether they were experiencing a “current significant episode of pain” in the last month (0 = no; 1 = yes), identify the location(s) of the pain episode on a body map, rank the pain severity (from 0 = no pain; to 100 = worst pain imaginable), describe the quality of the pain episode (tingling, stabbing, etc), and report the age they were when the pain began. Pain severity rating scales are commonly used to measure pain intensity and show moderate-to-high reliability (0.67–0.96).³⁰ The current age of the participants in addition to their self-report of when pain began was used to construct a pain duration measure.

This self-report of a pain episode includes multiple pain sites (eg, head pain and back pain) and is best interpretable as general chronic pain (see **Table 1** for pain sites). Chronic pain inception can begin at an individual location but often spreads to multiple locations, which appears evident in our sample where 54% of those in pain are endorsing multisite pain. It is theorized that the spreading of pain may occur due to central sensitization, which is an amplification of pain due to the maladaptive plasticity of the central nervous system.^{27,56} While chronic pain conditions are surely diverse in causes and symptoms, there is both psychometric and genetic evidence that there is a shared etiology for pain across body sites. A common component underlying pain starting at a single site and spreading across body sites has been longitudinally validated.⁷ Furthermore, both twin⁵⁷ and molecular genetic⁵⁹ studies have identified a common genetic factor that underlies all measured chronic pain conditions; thus, looking at general chronic pain may be useful for understanding part of the pathophysiology.

Although the BPHQ is self-reported, we attempted to ascertain a measure of pain chronicity as closely related to clinical cut-offs as possible. For these analyses, no pain was ascertained by answering “No” for experiencing a significant episode of pain (no chronic pain = 0). Participants who were experiencing non-chronified pain that was mild (mild chronic pain; pain intensity ≤ 29) and/or persisted for <3 months were coded as 1. Participants whose pain had lasted for 3+ months and was moderate to severe in intensity (moderate–severe chronic pain; pain intensity ≥ 30) were coded as 2. The optimal cut-offs for pain severity may vary, but generally a cut-off around 34 can be used for mild pain.¹¹ We chose the cut-off of 30 as this has been conceptualized as clinically meaningful pain by a previous prospective EF and pain study.³ Creating a pain variable using a combination of pain severity and duration has been used before in relation to EF⁴² and is meant to ascertain a measure of chronic, debilitating pain.

Of the 637 people who filled out the BPHQ questionnaire, one person who reported pain rated their pain severity as 0, and 2 people with pain were missing on their pain severity rating; hence, our final *n* for the pain measure was 634. Of the people who reported having pain in our sample, 97.2% reported pain lasting more than 3 months, making our measure interpretable as general chronic pain. We decided not to exclude the 5 participants whose pain episode was <3 months as their pain

Table 1
Frequency of self-reported pain locations.

Pain locations	
Single body location	43%
Multiple body locations	54%
Lower extremity/Knee	35%
Back	27%
Upper extremity/Shoulder	13%
Head/Neck	13%
Chest	3%
Stomach/Bowel	3%
Hip	3%
Genital	2%
Chose not to report	3%

The percentages above are calculated from the 176 people who reported experiencing chronic pain. Over half of the sample was experiencing pain in multiple locations.

episodes were showing persistence (range = 1.18–2.37 months), and our models would then lose information about their co-twins as well. To ensure these 5 individuals were not changing results, we ran a version of the phenotypic Cholesky decomposition in which we dropped them entirely. The results were consistent. While we do not have a measure of pain history at age 23, the average pain episode duration reported at age 28 was 1.88 years (22.60 months) suggesting that most of the participants' pain inception occurred after age 23.

2.2.2. EF tasks

Participants completed 9 computerized EF tasks at the age 23 visit (antisaccade, stop-signal, Stroop, keep-track, letter memory, spatial *n*-back, number-letter, category-switch, and color-shape). At age 28, the battery was shortened to 6 tasks (antisaccade, Stroop, keep-track, letter memory, number-letter, and category-switch). The tasks are briefly described below (see Refs. 23 and 26 for more details).

The 3 response inhibition tasks required participants to avoid prepotent responses. In the *antisaccade task*, participants had to avoid the reflex to look at a cue stimulus on one side of the screen and instead immediately look in the opposite direction to see a briefly appearing target stimulus (a number from 1 to 9). The timing of the task was such that it would not be possible for the participants to see the target before it was masked if they saccaded to the cue. The dependent measure was the accuracy of target identification. In the *stop-signal task*, participants had to categorize an arrow as pointing right or left as quickly and accurately as possible. In 25% of the trials, the participants had to withhold their categorization if the arrow on the screen turned red (the delay before the arrow turned red was titrated according to performance). The dependent measure was the stop-signal reaction time. The stop-signal task was not collected at age 28. In the *Stroop task*, the participants had to avoid the tendency to read color words (eg, red, blue, green) and instead report the font color. There was a block of 42 trials of colored strings of 3 to 5 asterisks; then, a block of 42 trials of color words printed in a congruent color; then, 2 blocks of 42 trials of color words printed in an incongruent color. The difference in mean reaction times for correct responses for the incongruent and asterisk blocks was the dependent measure.

The 3 updating working memory tasks required participants to add and delete information in working memory. In the *keep-track task*, participants had to keep track of the last word appearing in 2 to 5 categories from a series of words from 6 different categories (animals, colors, countries, distances, metals, and relatives). For each trial, a category list was shown and remained at the bottom of the screen while 15 to 25 words appeared serially for 2 seconds each. The proportion of target words recalled across all trials was the dependent measure. In the *letter memory task*, participants had to rehearse the most recent 4 letters from a series that were 9, 11, or 13 letters long. On each trial, consonants appeared one at a time for 3 seconds each, during which time participants would say aloud the last 4 letters including the letter currently on the screen. The dependent measure was the proportion of sets correctly rehearsed across 12 trials (132 sets total). In the *spatial n-back task*, participants saw 12 squares fixed in random locations on the screen. Within 6 blocks, 24 squares flashed one at a time, and the participants had to indicate whether it was the same square that flashed either 2 or 3 trials before. The 2-back and 3-back were separate conditions, and participants completed 6 blocks for each. The dependent measure was the averaged *z*-scores of the arcsined proportion of correct scores for

both conditions. The spatial *n*-back task was not administered at age 28.

The 3 set-shifting tasks required participants to switch between 2 subtasks based on a cue that appeared 350 ms before the stimuli in each trial. In each trial of the *number-letter* task, participants saw a number-letter/letter-number pair in one of 4 quadrants of a square. They categorized the number as odd or even when the pair appeared in one of the 2 top quadrants and categorized the letter as consonant or vowel if the pair appeared in one of the 2 bottom quadrants. In each trial of the *color-shape* task, the participants saw a colored shape and categorized the color as red or green when it appeared with a "C" cue or categorized the shape as a circle or triangle when it appeared with an "S" cue. The *color-shape* task was not administered at age 28. In each trial of the *category-switch* task, participants saw a word and categorized it as either being smaller or bigger than a soccer ball or living or nonliving depending on which cue symbol appeared. In all 3 tasks, half the trials required repeating the subtask from the prior trial and the other half required shifting subtasks. The dependent measure for all the switch tasks was the difference between the average reaction time for switch trials and the reaction time for repeat trials.

2.3. Study design

The design of this study is an observational individual twin differences twin study (ie, no experimental manipulation). Our sample was not selected for chronic pain nor for EF performance levels. We used the longitudinal, observational data of our sample to run cross-sectional and longitudinal structural equation models. We used the twin structure of our sample to run genetic models.

2.4. Statistical procedures

We used Mplus Version 8.10⁴³ to estimate both the phenotypic and genetic models. The analyses were conducted on raw data, and the models used all available data (see **Table 2** for the *n* for each variable). The EF models consisted of continuous data and were estimated using the robust maximum likelihood estimator. The models that included the ordinal pain measure were estimated using the diagonally weighted least squares mean and variances adjusted estimator. Mplus' TYPE = COMPLEX option was used for the phenotypic models, which employs a sandwich estimator to adjust SEs and χ^2 tests to correct for the nonindependence due to the twin structure.⁴⁸ A non-significant χ^2 indicates that the model is perfectly reproducing the observed covariance matrix; however, due to the χ^2 's sensitivity to sample size, a significant χ^2 does not necessarily indicate an unreasonable fit (it indicates a nonperfect fit). Therefore, we used the comparative fit index (CFI) > 0.95 and root mean square error approximation (RMSEA) < 0.06 to indicate a good fit.²⁸ A significant parameter was assessed at $P < 0.05$ for the Wald tests of the phenotypic parameters. The significance of the variance components of the genetic models was assessed using Mplus's diffest procedure, as the SEs in the genetic models are not invariant to model parameterization, but the χ^2 difference test is.⁴⁴

2.4.1. Data transformation

Sex was included as a covariate of pain in all analyses, and age and sex effects were regressed out of the EF tasks at each respective time point. Age was not included as a covariate of pain

Table 2
Descriptive statistics for executive function tasks.

	n	Mean	SD	Min	Max	Skew	Kurtosis	Reliability
EF age 23								
Antisaccade	748	0.62	0.16	0.20	0.96	−0.13	−0.67	0.90*
Stop-signal	735	215 ms	30	116	315	−0.23	0.25	0.63*
Stroop	737	156 ms	74	−73	387	0.71	0.71	0.96†
Keep-track	749	0.72	0.09	0.44	0.96	−0.36	0.11	0.66*
Letter memory	749	0.70	0.13	0.38	1.00	0.22	−0.64	0.92*
Spatial <i>n</i> -back	749	−0.01	0.91	−2.74	2.70	−0.31	−0.03	0.75†
Number-letter	748	246 ms	157	−241	735	0.91	0.92	0.91†
Color-shape	743	221 ms	182	−239	792	1.05	1.19	0.90†
Category-switch	747	198 ms	161	−81	735	1.14	1.28	0.94†
EF age 28								
Antisaccade‡	613	0.00	1	−2.36	2.48	−0.04	−0.74	0.90*
Stroop	632	147 ms	78	−0.90	385	0.70	0.73	0.96†
Keep-track	629	0.77	0.14	0.35	1.00	−0.69	0.20	0.75*
Letter memory	637	0.73	0.15	0.31	1.00	−0.05	−0.87	0.93*
Number-letter	616	183 ms	121	−60	566	0.89	0.82	0.92†
Category-switch	636	180 ms	150	−103	662	1.19	1.45	0.97†

Descriptive statistics of the executive function (EF) tasks.

* Internal reliability was calculated using Cronbach's α .

† Internal reliability was calculated by using split-half reliability corrected with the Spearman-Brown prophecy formula.

‡ The Antisaccade task at age 28 was a *z*-score combination of 2 difficulty levels, so these descriptive statistics reflect the standardized score.

as it was a nonsignificant predictor estimated at near 0 ($r = 0.00$ [0.01], $P = 0.996$). The cohort showed a narrow age range, and the testing protocol attempted to minimize age variance. All the EF tasks were rescaled to have an SD of around 1 to help speed model estimation.³²

2.4.2. EF factors

The EF model we used is based on previously published analyses of these EF data (Fig. 1).²³ The factors reflect a unity and diversity model of EFs, which has been well supported in the literature.²⁴ The Common EF captures variance common to all the EF tasks and is thought to reflect general goal maintenance or top-down biasing of attention.³⁹ The Updating-specific factor captures the left-over shared variance of the 3 updating working memory tasks after accounting for Common EF variance. Working memory is a short-term information retention memory system that maintains features of importance, filters out unnecessary sensory signals,

and continuously adds and deletes information.⁴ Similarly, the Shifting-specific factor captures variance shared by the 3 shifting tasks that is not accounted for by the Common EF factor. There is no factor specific to the inhibiting tasks because the Common EF factor captures all the variance shared by the 3 response inhibition tasks.

This EF model was also applied to the age 28 data based on a previous study using these data.⁵⁴ Unlike at age 23, the stop-signal, spatial *n*-back, and color-shape tasks were not included in the age 28 battery. Since the age 28 Updating-specific and Shifting-specific factors were left with only 2 indicators each, the loadings were equated (after scaling the tasks to have similar variances) to allow for identification.³¹ For longitudinal models, residual covariances between the same task across time were included. At age 23, a residual covariance between the antisaccade and spatial *n*-back task was included, as prior studies using these EF data have reported the model with this shared measurement error.²⁵

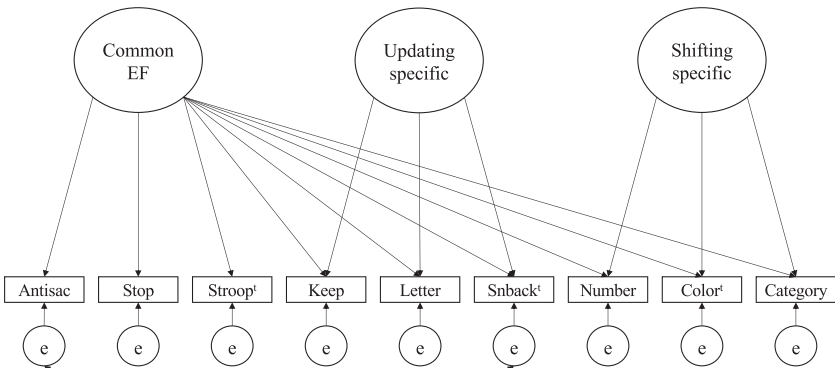


Figure 1. Executive function (EF) model, specified based on prior publications using these EF data.^{25, 49} The Common EF factor captures shared variance across all the EF tasks. The Updating-specific factor captures shared variance specific to the working memory updating tasks and the Shifting-specific factor captures shared variance specific to mental set shifting, both after the variance attributable to Common EF is considered. Each task is allowed an error term that includes the measurement error and variance unique to each task not explained by the 3 factors. The antisaccade and spatial *n*-back share a residual correlation possibly because both involve rapid eye movements to different spatial locations on the screen. This residual correlation has been included in all publications that use these data as it improves model fit and aids in convergence for the genetic models, see Friedman et al.²⁵ for details. These 3 tasks were not collected at age 28 and consequently not included in the age 28 model of EF. Antisac, antisaccade; Color, color-shape; Category, category-switch; Keep, keep-track; Letter, letter memory; Number, number-letter; Snback, spatial *n*-back; Stop, stop-signal.

2.4.3. Phenotypic longitudinal Cholesky decomposition

We used a phenotypic Cholesky decomposition to evaluate whether it is the stable variance components of the EF factors (variance at age 23 and the variance it shares with age 28) or the new EF variances that appear concurrently with pain at age 28 that relate to pain at age 28. A Cholesky decomposition is a useful technique for longitudinal data as it extracts all variance from the second time point of a variable that is shared with the first time point and partitions the residual variance of the second time point into a separate factor.⁵ Thus, it allows for the examination of partial correlations with the second time point, controlling for the earlier time point.

Previous research using an earlier wave at age 17 and the age 23 wave of this sample has found that EF factors are highly stable across time.²⁵ Thus, we decided to use a Cholesky decomposition of the age 23 and 28 waves: The first factor predicts all the variance at age 23 and the variance at age 28 that is shared with age 23; the second factor captures age 28 variance that is unique (Fig. 2). We correlated both the age 23 and the age 28–specific variance components of EFs with pain, for a total of 6 covariances with pain. We did so to assess whether the pain is related to the stable components of EF, which existed 6 years before the pain report, or whether it is related to EF variance specifically at age 28, concurrent with the time of the pain report.

2.4.4. Genetic models

We employed a series of genetic models to decompose the variances and covariances of pain and EFs into genetic and environmental influences.⁵¹ The classical ACE twin model partitions phenotypic variance into additive genetic (A), shared environmental (C), and nonshared environmental (E) variances. Since MZ twins share 100%, but DZ twins share on average 50%, of their segregating genes, their A components correlate at 1.00 and 0.50, respectively. Twins who are reared in the same environment are assumed to correlate 1.00 on the C component (eg, socioeconomic status, nutrition). The E component is uncorrelated for both zygositys as it represents unique influences (eg, injury of one twin) and includes measurement error for nonlatent factors. To select the most parsimonious model, the individual components of a model can be dropped (A or C) and tested with a χ^2 difference test. If dropping a variance component does not significantly affect model fit, the most parsimonious model is used (eg, if the C contributes 0% in an ACE model, an AE model will be selected as the best fitting model).

Cholesky decompositions are also often applied to twin models with longitudinal data to partition the genetic (A) and environmental (C, E) variances into variance in the second time point that is shared with the first time point and unique variance that has newly emerged at the second time point.³⁵ We first

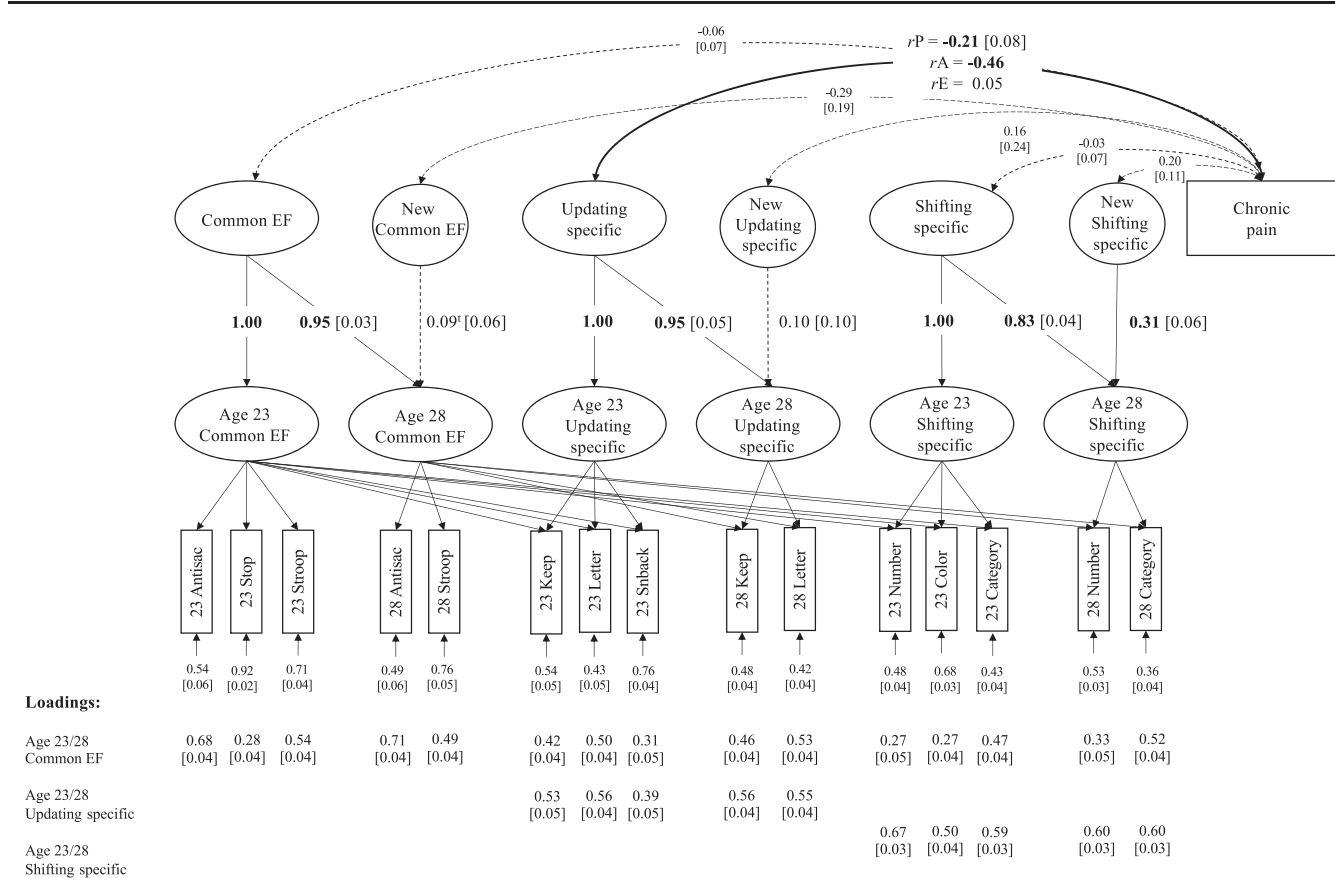


Figure 2. Estimates from Cholesky decomposition of executive function (EF) factors and their associations with pain. Only the Shifting-specific factor had significant residual variance at age 28; however, we phenotypically correlated pain with all 3 stable factors and all 3 residual factors. Pain significantly correlated with the stable Updating-specific variance, but not the stable Common EF nor Shifting-specific variances. Pain did not significantly associate with any of the age 28–specific EF variances. We display the genetic and environmental decomposition of the phenotypic Updating-specific and pain correlation estimated in the genetic Cholesky decomposition. The loadings of each task on its respective factors are shown. SEs are in brackets. Bold font indicates $P < 0.05$, based on Wald estimates in the phenotypic model, and χ^2 difference tests in the genetic model. *Significance at a $P < 0.10$ level. 23, age 23; 28, age 28; Antisac, antisaccade; Color, color-shape; Category, category-switch; Keep, keep-track; Letter, letter memory; Number, number-letter; rP, phenotypic correlation; rA, genetic correlation estimated from the genetic Cholesky decomposition; rE, nonshared environmental correlation estimated from the genetic Cholesky decomposition; Snback, spatial n-back; Stop, stop-signal.

estimated a genetic and environmental Cholesky decomposition of just the EF factors to assess whether there were any new, significant A, C, or E influences for the EF factors that emerged at age 28. We estimated all possible cross-paths at both the factor level (eg, shared genetic variance of the Common EF factor at age 28 with age 23) and at the task-specific level (eg, shared genetic variance of antisaccade task at age 28 with age 28 not accounted for by the Common EF factor).

Based on the genetic EF model, we included the nonzero, significant variance components in our model with pain. As our pain variable is ordinal, it requires a diagonally weighted least squares estimator which sometimes has difficulty converging in large models with many parameters estimated at 0. In our genetic Cholesky of pain and EFs, we decomposed the phenotypically significant associations between pain and the EF factors and assessed whether they were genetic or environmental in nature. The estimates from this twin Cholesky decomposition were used to derive genetic (r_A) and nonshared environmental (r_E) correlations between EFs and pain.³⁵ For example, the r_A for Common EF and pain quantifies the overlap of genetic influences.⁵¹ We conducted power analyses for these models which can be found in the Supplementary Results (<http://links.lww.com/PAIN/C102>).

3. Results

3.1. Descriptive statistics

Descriptive statistics for the EF tasks are presented in **Table 2**, and zero-order correlations are presented in **Table 3**. Pain scores were available for 634 participants: n of no chronic pain = 458; n of mild chronic pain = 102; and n of moderate–severe chronic pain = 74. For the genetic models, the cross-tabulation of the number of twin pairs in each zygosity and chronic pain level is shown in Supplementary Table 1 (<http://links.lww.com/PAIN/C102>).

3.2. Phenotypic associations

The phenotypic model of correlations between age 28 pain and the age 23 EF factors fit well, $\chi^2(26) = 31.406$, $P = 0.214$, CFI = 0.993, RMSEA = 0.016, model $n = 785$. Pain significantly correlated with the age 23 Updating-specific factor ($r = -0.20$,

$P = 0.006$), but not with the Common EF ($r = -0.06$, $P = 0.349$) nor Shifting-specific ($r = -0.03$, $P = 0.733$) factors. The phenotypic model of the age 28 EF factor correlations with pain fit acceptably, $\chi^2(10) = 43.440$, $P < 0.001$, CFI = 0.923, RMSEA = 0.072, model $n = 649$. Pain significantly correlated with the age 28 Common EF factor ($r = -0.15$, $P = 0.015$), marginally with the Updating-specific factor ($r = -0.15$, $P = 0.061$), but nonsignificantly with the Shifting-specific factor ($r = 0.09$, $P = 0.228$).

We conducted descriptive phenotypic correlations between pain and the EF tasks to inform our interpretation of the full model (**Table 3**). Consistent with significant and marginally significant correlations of pain with the Updating-specific factors at ages 23 and 28, respectively, across both time points, pain had significant associations with the keep-track and letter memory tasks, which load onto the Common EF and Updating-specific factors. At age 23, the third highest association was with the n -back, which is also an Updating-specific task, but this did not reach significance. At age 28, the associations of the antisaccade and Stroop tasks with pain were approaching significance, which likely contributed to the significant association between pain and Common EF at this age.

3.3. Phenotypic Cholesky decomposition

The phenotypic Cholesky decomposition is visualized in **Figure 2**. The phenotypic Cholesky decomposition of the EF factors fit well, $\chi^2(81) = 143.598$, $P < 0.001$, CFI = 0.965, RMSEA = 0.031, model $n = 786$. There was significant residual variance specific to age 28 Shifting-specific factor (residual = 0.31, $P < 0.001$), but the residual variance for age 28 Common EF (residual = 0.09, $P = 0.097$) and Updating-specific (residual = 0.10, $P = 0.275$) factors was not significant.

Pain significantly correlated with stable Updating-specific variance ($r = -0.21$, $P = 0.008$) but was not significantly related to stable Common EF variance ($r = -0.06$, $P = 0.350$) nor stable Shifting-specific variance ($r = -0.03$, $P = 0.709$). Pain did not significantly relate to new Common EF ($r = -0.29$, $P = 0.142$), Updating-specific ($r = 0.16$, $P = 0.506$), nor Shifting-specific variance at age 28 ($r = 0.20$, $P = 0.063$).

Table 3
Zero-order correlations between pain and executive function tasks.

	Pain	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Age 23															
Antisaccade	−0.02 (0.05)	—													
Stop	0.03 (0.05)	−0.22	—												
Stroop	0.05 (0.05)	−0.32	0.13	—											
Keep	−0.14 (0.06)	0.25	−0.12	−0.22	—										
Letter	−0.17 (0.05)	0.40	−0.10	−0.28	0.50	—									
Spatial n -back	−0.09 (0.05)	0.36	−0.04	−0.18	0.32	0.40	—								
Number	0.03 (0.06)	−0.27	0.08	0.18	−0.11	−0.09	−0.04	—							
Color	0.03 (0.05)	−0.19	0.02	0.16	−0.14	−0.11	−0.10	0.43	—						
Category	0.05 (0.06)	−0.34	0.14	0.29	−0.22	−0.19	−0.14	0.51	0.42	—					
Age 28															
Antisaccade	−0.10* (0.05)	0.65	−0.20	−0.30	0.29	0.41	0.28	−0.20	−0.14	−0.28	—				
Stroop	0.09* (0.05)	−0.24	0.14	0.49	−0.21	−0.27	−0.14	0.15	0.13	0.26	−0.25	—			
Keep	−0.14 (0.05)	0.29	−0.13	−0.25	0.53	0.50	0.32	−0.06	−0.17	−0.19	0.38	−0.25	—		
Letter	−0.17 (0.05)	0.37	−0.11	−0.31	0.51	0.84	0.37	−0.08	−0.14	−0.19	0.44	−0.30	0.55	—	
Number	0.02 (0.05)	−0.25	0.16	0.15	−0.10	−0.07	−0.07	0.55	0.31	0.47	−0.34	0.22	−0.11	−0.11	—
Category	0.00 (0.05)	−0.34	0.20	0.31	−0.25	−0.23	−0.13	0.47	0.38	0.66	−0.34	0.28	−0.22	−0.23	0.54

Pain was assessed at age 28. Executive function tasks were completed longitudinally at ages 23 and 28 years. Age and sex effects were regressed out of the executive function tasks. Sex was included as a covariate for pain. Age was not included as a covariate for pain since it had a nonsignificant association estimated at 0. The SEs for correlations with pain are shown in brackets, corrected for the nonindependence of twin pairs. Bolded font indicates $P < 0.05$, accounting for nonindependence.

* $P < 0.10$.

3.4. Genetic models

The MZ twin correlation for pain ($r = 0.46, P < 0.001$) was over twice as large as the DZ twin correlation for pain ($r = 0.12, P = 0.368$), which is indicative of additive genetic effects (A) but no shared environmental effects (C). A univariate AE twin model for this pain measure indicated significant $A = 41\%$, $\chi^2(1) = 16.902, P < 0.001$, and $E = 59\%$, $\chi^2(1) = 45.410, P < 0.001$, influences (the latter include measurement error).

Before adding pain into the EF model, a genetic and environmental Cholesky decomposition of EFs to assess whether there was significant stable and/or new ACE variance (Fig. 3). All possible cross-paths were included at the EF factor level as well as at the task-specific level (see task-specific ACE estimates in Supplementary Table 2, <http://links.lww.com/PAIN/C102>). We used the path estimates to calculate the genetic and environmental correlations from age 23 to age 28 EFs. The model fits well, $\chi^2(867) = 1109.173, P < 0.001$, RMSEA = 0.037, CFI = 0.945, model $n = 402$. The residual paths suggested there was no new significant A or C variance unique to age 28 for any of the EF factors. There was no significant unique E variance for Common EF (0.04, $\chi^2(1) = 2.104, P = 0.147$) nor for the Updating-specific factor (0.06, $\chi^2(1) = 0.015, P = 0.903$) at age 28 (Fig. 3). The only significant new variance that emerged at age 28 was unique E variance for the Shifting-specific factor (0.23, $\chi^2(1) = 8.521, P = 0.004$). Hence, genetic variance at age 28 was entirely shared with age 23 genetic variance for Common EF ($rA =$

1.00, $\chi^2(1) = 46.651, P < 0.001$), Updating-specific ($rA = 1.00, \chi^2(1) = 27.539, P < 0.001$), and Shifting-specific ($rA = 1.00, \chi^2(1) = 15.569, P < 0.001$) factors. The shared environmental variance was nonsignificant for all 3 factors and was estimated at 0 or near 0. The Common EF factor shared significantly stable E variance across time ($rE = 0.74, \chi^2(1) = 9.671, P = 0.002$). The nonshared environmental variance was estimated near 0 and not significant for the Updating-specific factor at age 23, and hence it could not share stable E variance with age 28 ($rE = 0.64, \chi^2(1) = 0.155, P = 0.694$). Nonshared environmental variance at age 28 did not share significantly stable nonshared environmental influences with the age 23 Shifting-specific factor ($rE = 0.40, \chi^2(1) = 2.533, P = 0.111$).

Based on the phenotypic Cholesky decomposition of pain and EFs and the genetic and environmental EF Cholesky decomposition, we estimated a genetic and environmental Cholesky decomposition for pain and EFs including cross-paths to all phenotypically significant associations (see model fit statistics in Table 4). The only significant phenotypic correlation was with the Updating-specific factor; hence, this was the only association we decomposed (but for thoroughness, we present decomposition fit statistics for a model that includes all cross-paths with nonsignificant phenotypic associations and nonsignificant variance components in Supplementary Table 3, <http://links.lww.com/PAIN/C102>). The model fit well, $\chi^2(1083) = 1122.372, P = 0.198$, RMSEA = 0.013, CFI = 0.984, model $n = 403$. Pain genetically correlated with the stable genetic component of the

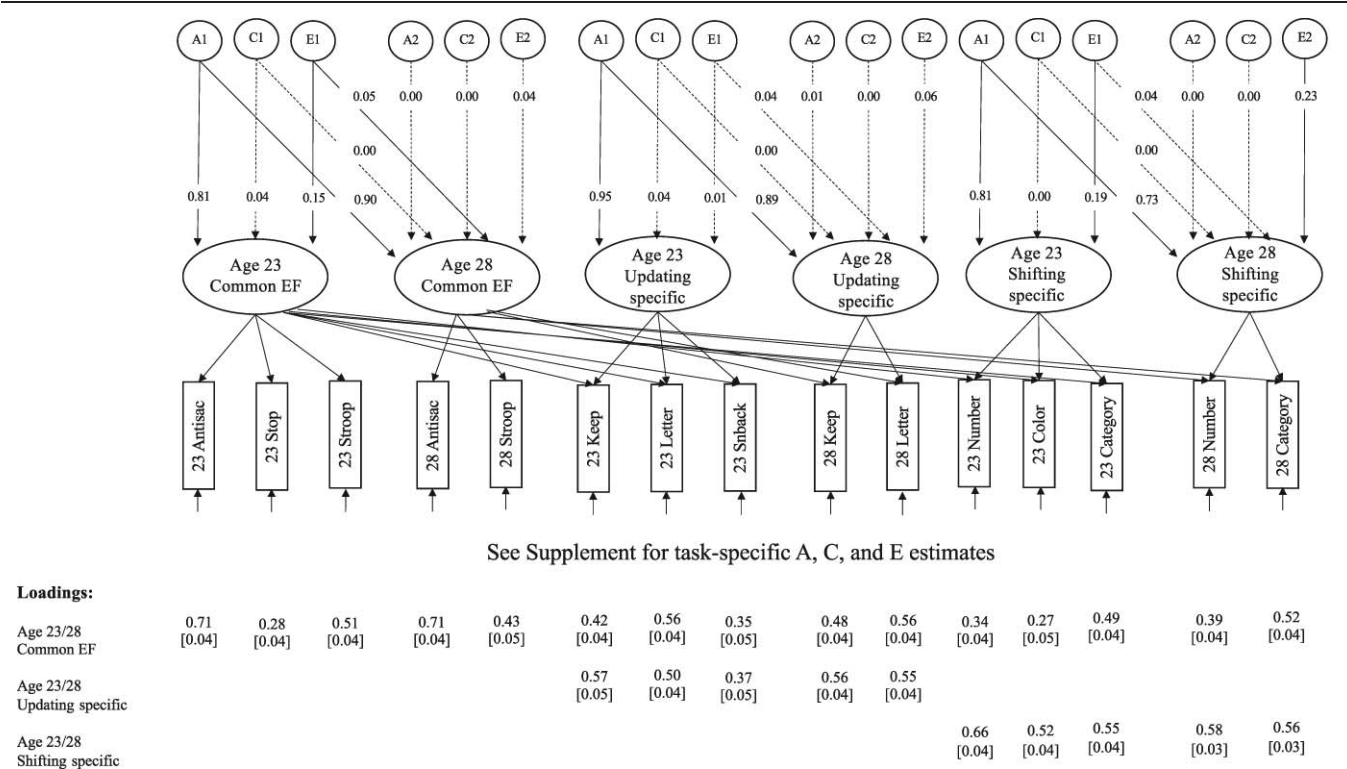


Figure 3. Estimates from the genetic Cholesky decomposition of executive function (EF) factors. The A1s have direct paths to the age 23 factors as well as a cross-path to the age 28 factors. Hence, A1 captures genetic variance in the age 23 factor and the variance it shares with the age 28 factor. A2 has a direct path to the age 28 factor and is the residual genetic variance specific to the age 28 factor. This decomposition was done for additive genetic (A1, A2), shared environmental (C1, C2), and nonshared environmental (E1, E2) variances for each EF factor (ie, Common EF, Updating-specific, and Shifting-specific). In this Cholesky decomposition, the tasks also all had their own task-specific ACE decompositions (eg, an A1 for age 23 antisaccade and age 28 antisaccade and an A2 for age 28 antisaccade). These task-specific estimates are not visualized here for space but can be found in Supplementary Table 2 (<http://links.lww.com/PAIN/C102>). The solid lines symbolize significance at $P < 0.05$; dashed lines indicate $P > 0.05$, determined with χ^2 difference tests (the A1, C1, and E1 paths cannot be tested with single- df difference tests when the age 28 variables are included, as dropping the first path requires also dropping the cross-paths to ensure model identification; therefore, significance tests for the age 23 paths were obtained from Friedman et al.,²⁵ which presented the age 23 model). 23, age 23; 28, age 28; A, additive genetic effects; C, shared environmental effects; E, nonshared environmental effects (includes measurement error); Antisac, antisaccade; Color, color-shape; Category, category-switch; Keep, keep-track; Letter, letter memory; Number, number-letter; Stop, stop-signal; Sback, spatial n -back.

Table 4
Fit statistics for the Cholesky decomposition of the Updating-specific association with pain.

Model	Model fit					χ^2 difference	
	χ^2	Df	P	RMSEA	CFI	$\Delta\chi^2(1)$	P
Full model	1122.372	1083	0.198	0.013	0.984		
Drop stable A Updating-specific association	1132.240	1084	0.150	0.015	0.980	7.879	0.005
Drop stable E Updating-specific association	1123.021	1084	0.200	0.013	0.984	0.039	0.844

Presented above are the fit statistics of the genetic Cholesky decomposition between pain and executive function (EF). We only decomposed significant phenotypic associations between pain and EF, which was the association between pain and the stable Updating-specific factor. A significant worsening of model fit when dropping an association from the model suggests that the association is statistically significant. Pain shares a significant genetic association with Updating-specific factor, but not an environmental association. Sex was included as a covariate of pain; age and sex effects were regressed out of the EF tasks.

Bolded font indicates $P < 0.05$, determined with χ^2 difference tests.

A, additive genetic effects; CFI, comparative fit index; E, nonshared environmental effects (includes measurement error); df, degrees of freedom; RMSEA, root mean square error approximation.

Updating-specific factor ($rA = -0.46$, $\chi^2(1) = 7.879$, $P = 0.005$). Pain did not share significant nonshared environmental correlations with the stable Updating E component ($rE = 0.05$, $\chi^2(1) = 0.039$, $P = 0.844$).

4. Discussion

To our knowledge, this is the first study to examine the phenotypic, longitudinal, and genetic/environmental relationships between chronic pain and 3 separable EF latent variables (Common EF, Updating-specific, and Shifting-specific factors). We found that chronic pain in young adults is associated with lower scores on a factor specific to working memory updating and that this relationship is entirely due to shared genetic risk.

Worsened working memory Updating-specific function predicting pain is not surprising. Working memory updating is highly correlated with more general working memory factors based on complex working memory span tasks.⁵³ Meta-analyses and systematic reviews have found that across multiple pain syndromes and ages, patients with chronic pain tend to perform worse on working memory tasks.^{9,13} Clinical pain patients also self-report pronounced working memory deficits compared with pain-free controls.⁶ Preoperative pain-free patients who performed worse on processing speed tasks were at higher risk for developing chronic pain 6 and 12 months postoperatively; however, the tasks used also tapped into cognitive flexibility.³ Our findings suggest that the impairments are specific to working memory updating, as variance shared with response inhibition and set-shifting impairments were controlled for by the other latent variables.

The lack of significant correlations between pain and the Common EF and Shifting-specific factors differs from what has been reported in studies that used single EF tasks. Multiple studies have found that patients with chronic pain perform worse on individual inhibition and shifting tasks,^{8,13} although our findings align with a study of Japanese older adults that found that individuals with musculoskeletal pain showed slower processing speed but no significant differences on a shifting task⁴² compared with controls. A strength of our research is its large sample size and the use of latent variable modeling, which minimizes measurement errors common in cognitive tasks. Including the multiple EF latent variables, which partition the effects of different EF domains, we were able to isolate that chronic pain is related to Updating-specific working memory over and above Common EF and Shifting-specific ability. It is possible that shared variance with working memory in the inhibition and shifting tasks has driven associations in other research. Alternatively, Updating-specific impairments may be related specifically to mechanisms of general chronic pain, such as central sensitization.²⁷

Updating-specific function prospectively predicting pain suggests working memory-specific deficits could play a role in pain chronification. A Midlife Development in the United States study (average age = 54.43) constructed a mean z-score of 5 EF tasks tapping inhibition, working memory, and set shifting and found that baseline EF predicted the development of chronic pain 9 years later, even after adjusting for depressive symptoms, personality, age, and sex.⁴⁵ A 12-year longitudinal study using an older adult cohort (average age = 77.46) found that pain status was related to impairment on a global latent factor of multiple EF tasks.⁵² Our results expand the finding of a factor-level cognitive impairment to a young adult sample and isolate that it is Updating-specific functioning that relates to pain chronification.

We used behavior genetic methods to understand what risk factors were contributing to the co-occurrence of pain and the Updating-specific factor. When decomposing the pain–Updating-specific factor association, we found that the relationship was entirely genetic. Variation in Updating-specific variance is almost 100% accounted for by additive genetic effects,²⁵ and general chronic pain is usually over 50% attributable to genetic effects across time.¹⁴ Shared genetic risk can be interpreted in multiple ways, but can suggest that the same or proximal risk variants are causally impacting pain and working memory, possibly through shared biological pathways.³⁷ While this study cannot elucidate biological pathways, our finding of a genetic correlation supports the conclusion that the pain–EF relationship is due to inherited factors and not directly due to environmental influences, such as a stressful life event over which the individual has no control. However, gene–environment correlations, in which genetic predisposition drives exposure to certain environmental influences (eg, engaging in risky behavior), or genotype-by-environment interactions, in which genetic risk genotypes may be more influential in some environmental contexts (eg, with greater exposure to air pollution), are still plausible.^{18,46}

More generally, the genetic relationship between pain and the Updating-specific factor is notable, as the only other known phenotype related to the Updating-specific factor is intelligence, and this relationship occurs in addition to a relationship with Common EF.³⁹ Thus, the pain–Updating-specific relationship is the only known association in which the Updating-specific factor shares a relationship with a phenotype exclusive of the other EF factors. Previous research has found that the processes that contribute to both working memory and pain perception seem to share the same limited cognitive resources,¹² which is consistent with our findings.

There are multiple mechanisms by which working memory updating could be impaired in the pain population. Working memory enables individuals to maintain, manipulate, and update (gate) relevant information, particularly in the context of

interference.^{5,21} One possibility is that the working memory system inefficiently gates ascending pain signals. Failure to filter out irrelevant sensory information could enhance pain perception in those suffering from chronic pain. Another possibility is that dispositional hypervigilance to threat cues or fear of pain may lead working memory to selectively monitor for pain signals and inappropriately update pain-related information. Anxiousness is related to poorer working memory performance⁴¹ and to pain³⁴; in addition, neuroticism shares genetic overlap with pain conditions.³⁷ However, EF has been shown to predict later pain over and above personality⁴⁵ and trait anxiety.³

In healthy individuals, the perception of pain depends on the attention that they allot to an experimental nociceptive stimulus, a process that may involve working memory.³³ Taxing working memory by engaging in a cognitively demanding task inhibited the perception of pain severity in experimentally induced pain across both younger and older adults.^{19,20} Transcranial magnetic stimulation of the dorsolateral prefrontal cortex (dlPFC), the cortical hub for working memory, during a working memory task also decreased the perception of pain severity.^{19,20} However, activation of the dlPFC without engagement in a cognitive task did not inhibit pain severity.²⁰ These findings suggest that, in healthy individuals, working memory may work with selective attention to help allocate limited cognitive resources and shield against nociceptive signals while maintaining attention on a current process. Genetic predisposition to deficiencies in this system could lead to improper filtering of pain and result in chronification.

Clinically, these findings support the usefulness of cognitive prioritization training to help minimize pain. Pain reprocessing therapy, which is a treatment focused on reevaluating the threat value of chronic pain, has been shown to provide pain relief in those with chronic back pain.² Furthermore, high-frequency transcranial magnetic stimulation over the dlPFC has been shown to decrease neuropathic pain for up to 3 months posttreatment.¹⁵

This study has a few limitations. At the second time point, 3 cognitive tasks were not included in the battery, making it difficult for the estimator to discriminate between Common EF and Updating-specific variance. Upon examination of the task-level correlations with pain, the only significant correlations were with the 2 updating tasks, and including the age 23 data made it clear the association was with the Updating-specific factor. However, the phenotypic correlations of pain with the Common EF tasks were marginally significant at age 28, hinting at possible emerging associations. Furthermore, the measure of pain was self-reported and the retrospective report of “when did your pain begin” is susceptible to bias.¹⁰ We do not have a measure of pain history at age 23, and while most participants reported their pain beginning after age 23, they could have a history of another pain episode. Finally, the results of this study may be limited to White individuals, as the LTS sample is not very diverse.

Given the novelty of these findings, there are many potential future directions to explore. First, this is a young adult sample, and it would be interesting to replicate these findings in an older adult sample, such as the United Kingdom Biobank, as an older population may have age-related cognitive declines and greater pain prevalence.¹⁷ Second, this study focused on general chronic pain. Individual pain conditions have their own unique risk that could lead to differential relationships with EF and differences in the underlying nature of the relationships. Third, once sample sizes become large enough to conduct a genome-wide association study of working memory, biological annotation of the genetic effects could elucidate what sorts of cellular mechanisms could be involved in working memory issues and subsequently pain chronification. Finally, while this study included

a robust battery of cognitive tasks, it by no means captures all the nuanced facets of EFs. Future research could benefit from looking at other factors of EF.

In conclusion, there is a shared genetic risk between worse working memory performance and pain in young adults. Issues with working memory could be mechanistically involved in pain chronification. An impaired gating mechanism of working memory or dispositional hypervigilance to pain cues could lead to overprioritization of pain signals.

Conflict of interest statement

The authors have no conflicts of interest to declare.

The data that support the findings of this study are available upon request.

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